Explaining Fibonacci periods

The field containing the roots controls the powers

$$\pi(11) = 10, \qquad \pi(7) = 16.$$

The structures we now understand

For a finite field K with q elements, $K^\times$ is cyclic of order $q-1$.

$$|\operatorname{SL}_2(\mathbb{F}_p)| = p(p^2-1).$$

For Fibonacci, with $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$,

$$\det A = -1, \qquad A^2 = A + I.$$

For odd p, apply Lagrange to $A^2 \in \operatorname{SL}_2(\mathbb{F}_p)$.

The Fibonacci period is the matrix order

For $t \geq 1$,

$$A^t = \begin{pmatrix} F_{t+1} & F_t \\ F_t & F_{t-1} \end{pmatrix}.$$

Since $F_{t-1} = F_{t+1} - F_t$,

$$A^t = I \iff (F_t, F_{t+1}) = (0, 1) \pmod{p}.$$

$$\pi(p) = \operatorname{ord}(A).$$

A first bound from SL_2

Let p be odd. If $A^t = I$, determinants give $(-1)^t = 1$.

Thus $\pi(p)$ is even, and

$$\mathrm{ord}(A^2) = \frac{\pi(p)}{2}.$$

Lagrange gives

$$\pi(p) \mid 2p(p^2 - 1).$$

Can the roots of $X^2 - X - 1$ give a sharper answer?

Choose a field containing the roots

Assume $p \neq 2, 5$.

- If 5 is square in $\mathbb{F}_p$, use $K = \mathbb{F}_p$.
- Otherwise, use $K = \{a + bs : s^2 = 5\}$ over $\mathbb{F}_p$.

$$\alpha = \frac{1+s}{2}, \qquad \beta = \frac{1-s}{2}.$$

$$\alpha + \beta = 1, \quad \alpha\beta = -1, \quad \alpha \neq \beta.$$

Diagonalize in the field where it makes sense

For either root r, since $r^2 = r + 1$,

$$A \begin{pmatrix} r \\ 1 \end{pmatrix} = r \begin{pmatrix} r \\ 1 \end{pmatrix}.$$

$$P = \begin{pmatrix} \alpha & \beta \\ 1 & 1 \end{pmatrix}, \qquad \det P = \alpha - \beta \neq 0.$$

$$P^{-1}AP = \operatorname{diag}(\alpha, \beta), \qquad A^t = I \iff \alpha^t = \beta^t = 1.$$

A period is a simultaneous order

We have shown

$$A^t = I \iff \alpha^t = \beta^t = 1.$$

If $\mathrm{ord}(\alpha) = r$ and $\mathrm{ord}(\beta) = s$, the return times are the common multiples of r and s.

$$\pi(p) = \mathrm{lcm}(r, s).$$

Case one: the roots are in $\mathbb{F}_p$

If 5 is a square in $\mathbb{F}_p$, both roots lie in $\mathbb{F}_p^\times$.

Fermat's theorem gives

$$\alpha^{p-1} = \beta^{p-1} = 1.$$

Therefore

$$A^{p-1} = I, \qquad \pi(p) \mid p-1.$$

Your turn: roots modulo eleven

In $\mathbb{F}_{11}$, use $s = 4$ and $2^{-1} = 6$.

1. Compute $\alpha = (1 + s)/2$ and $\beta = (1 - s)/2$.
2. Compute α^2 and α^5 by repeated squaring.
3. Which proper divisors of 10 could still be the period?

We already know that the period divides 10.

Modulo eleven: an exact period

Choose $s = 4$ in $\mathbb{F}_{11}$. Then

$$\alpha = 8, \qquad \beta = 4, \qquad \pi(11) \mid 10.$$

But

$$8^5 = -1, \qquad 8^2 = 9 \neq 1 \pmod{11}.$$

Every proper divisor of 10 divides 5 or 2.

$$\pi(11) = 10.$$

A Fibonacci root need not be primitive

Modulo 11, $\alpha = 8$ has order 10 and generates $\mathbb{F}_{11}^{\times}$.

Modulo 7, the roots live in $\mathbb{F}_{49}$, where

$$|\mathbb{F}_{49}^{\times}| = 48.$$

But $\alpha = 4 + 4s$ satisfies $\alpha^8 = -1$ and $\alpha^{16} = 1$.

$$\mathrm{ord}(\alpha) = 16, \quad \text{so it is not a multiplicative generator.}$$

A new symmetry: Frobenius

In characteristic p,

$$(u+v)^p = u^p + v^p.$$

The intermediate binomial coefficients are divisible by p.

Also $(uv)^p = u^p v^p$ and $1^p = 1$.

On a finite field, $u \mapsto u^p$ is injective, hence bijective:

$$u^p = v^p \quad \Longrightarrow \quad (u-v)^p = 0 \quad \Longrightarrow \quad u = v.$$

Which elements does Frobenius fix?

Every $a \in \mathbb{F}_p$ satisfies $a^p = a$.

The fixed elements are roots of

$$X^p - X.$$

A nonzero polynomial of degree p has at most p roots.

The prime field already supplies p roots.

$$u^p = u \iff u \in \mathbb{F}_p.$$

Case two: Frobenius exchanges the roots

Suppose 5 is nonsquare in $\mathbb{F}_p$, with $s^2 = 5$ in K.

$$(s^p)^2 = 5^p = 5.$$

Thus s^p is s or $-s$.

It cannot be s, because $s \notin \mathbb{F}_p$.

$$s^p = -s, \qquad \alpha^p = \beta, \qquad \beta^p = \alpha.$$

The minus sign tells us the return time

$$\alpha^{p+1} = \alpha\beta = -1, \qquad \beta^{p+1} = -1.$$

Diagonalization gives

$$A^{p+1} = -I, \qquad A^{2(p+1)} = I.$$

Therefore

$$\pi(p) \mid 2(p+1), \qquad \pi(p) \nmid p+1.$$

At time $p+1$, $(F_{p+1}, F_{p+2}) = (0, -1)$.

Modulo seven: the other exact period

Since 5 is nonsquare modulo 7,

$$A^8 = -I, \qquad A^{16} = I.$$

Thus $\pi(7) \mid 16$ but $\pi(7) \nmid 8$.

Every proper divisor of 16 divides 8.

$$\pi(7) = 16.$$

What the algebra explained

For primes $p \neq 2, 5$,

5 is square in $\mathbb{F}_p$	$\pi(p) \mid p-1$
5 is nonsquare in $\mathbb{F}_p$	$\pi(p) \mid 2(p+1)$

- Group size supplies a broad order bound.
- The field containing the roots supplies a sharper bound.
- Further power checks identify the least period.

Exit question

Suppose p is odd and 5 is nonsquare modulo p.

A student argues: "$F_{p+1} = 0$, so the period is $p + 1$."

What is missing, and what does the matrix actually tell us?

Use both entries of the Fibonacci state.

Why the prime five is different

Modulo 5, $X^2 - X - 1 = (X - 3)^2$.

$$A = 3I + N, \qquad N = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \qquad N^2 = 0.$$

For $t \geq 1$,

$$A^t = 3^t I + t3^{t-1} N.$$

$A^t = I$ requires $5 \mid t$ and $4 \mid t$.

$$\pi(5) = 20.$$

The prime two

Modulo 2, the successive Fibonacci states are

$$(0,1) \longmapsto (1,1) \longmapsto (1,0) \longmapsto (0,1).$$

$$\pi(2) = 3.$$

Here $-1 = 1$, and division by 2 is unavailable.

A bound need not be the least period

Modulo 29, $11^2 = 5$, so $\pi(29) \mid 28$.

The roots are $\alpha = 6$ and $\beta = 24$.

$$6^7 = -1, \quad 6^2 = 7 \neq 1; \qquad 24^7 = 1, \quad 24 \neq 1.$$

Their orders are 14 and 7.

$$\pi(29) = \operatorname{lcm}(14, 7) = 14.$$

An exact-period procedure

Start with a proven multiple B of $\pi(p)$.

1. Set $t = B$ and factor t.
2. For each prime divisor q of t, test $A^{t/q} = I$.
3. If the test succeeds, replace t by t/q and repeat.

When no prime factor can be removed, $t = \pi(p)$.